PAYTM BUILD FOR INDIA · TRACK 3

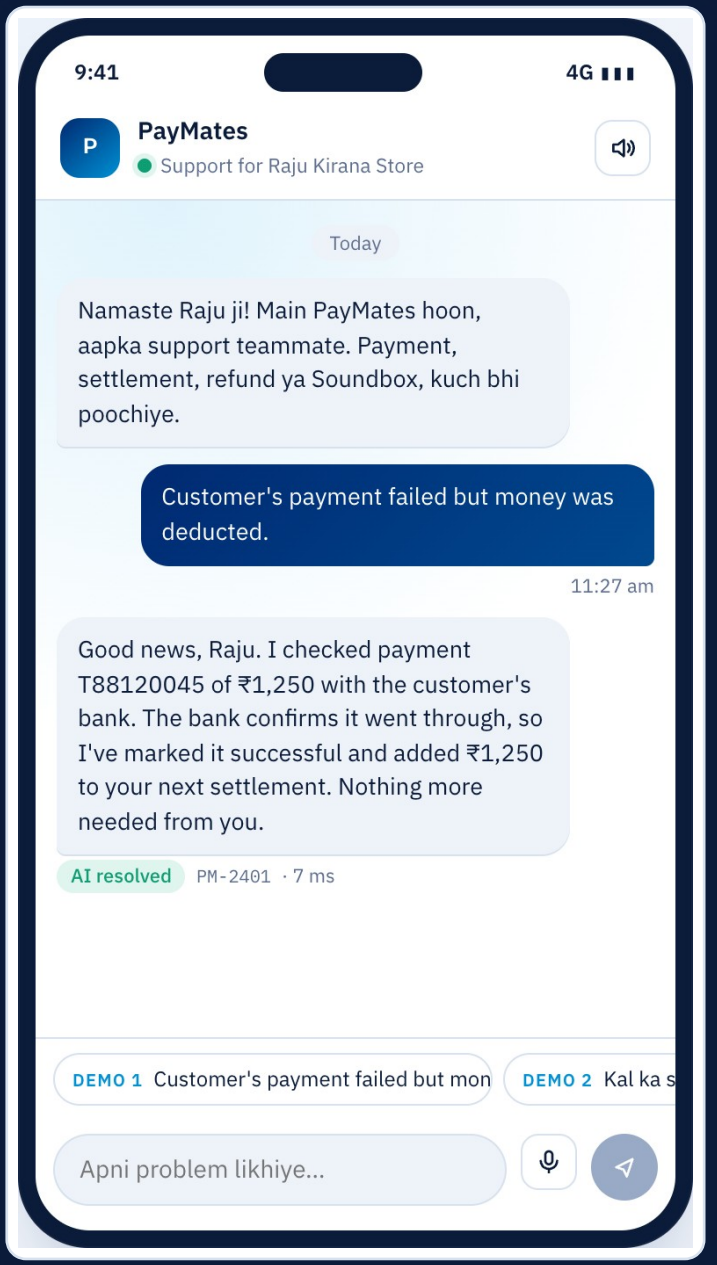
PayMates

The AI support teammate that fixes merchants' problems, not just answers them

English · हिन्दी · Hinglish

6 model providers

Model Council



At 9 PM, a stuck payment is a stuck business

Small merchants run on thin margins and tight cash flow. When money is stuck, they need it fixed now, not a ticket number and a callback.

~18 min

typical manual handling time per support ticket (our assumed baseline, labelled as an assumption in the app)

Failed payment, money deducted

The customer was debited, the merchant sees FAILED. Someone has to ask the bank.

Settlement didn’t arrive

Yesterday’s payout is on hold. The merchant can’t restock until it lands.

Refund bounced

The customer is angry and the merchant is stuck in the middle, with no update.

It doesn’t answer questions. It recovers money

DEMO 1 · ENGLISH

₹1,250

credited back

Bank rechecked, late success confirmed, payment credited to the next settlement.

DEMO 2 · HINGLISH

₹18,640

payout re-sent

Settlement held by a bank timeout, re-sent to the same verified account, ticket opened.

DEMO 3 · ENGLISH

₹7,800

held for a human

Bounced refund is above the ₹5,000 limit. Guardrail holds it; specialist approves in one click.

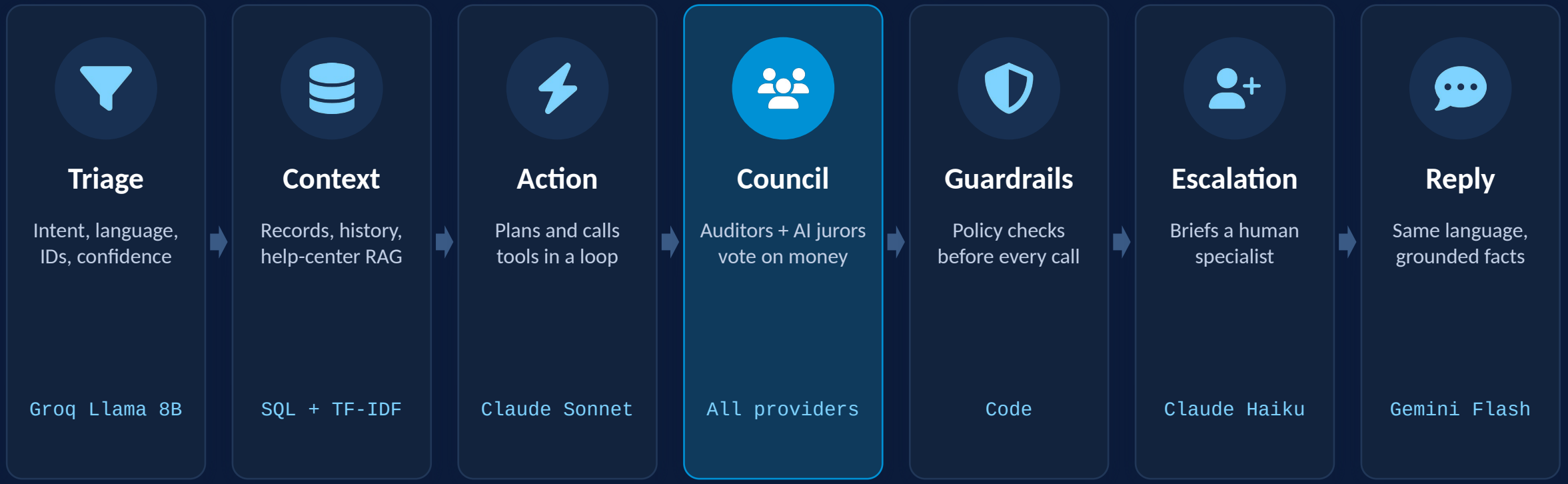
DEMO 4 · हिन्दी

₹3,400

council veto

Under the limit, but KYC is pending. The Model Council holds it and briefs a human.

A team of agents, one job each, streamed live



Every step is streamed to the UI over Server-Sent Events. If any model fails or times out, that step falls back to the rules engine and the ticket still closes.

The Model Council: no single model can move money

ACTION AGENT PROPOSES

Credit ₹3,400

T88340266



code · veto

Ledger auditor

Re-reads the evidence

code · veto

Replay auditor

Catches double payouts

code · veto

Velocity auditor

Risk-tiered daily cap

model · vote

Claude

Independent juror

model · vote

GPT

Independent juror

model · vote

Gemini · Llama

Independent jurors

Auditors can veto

Code checks the ledger, the audit log and a risk-tiered daily cap.

Jurors vote by majority

One juror per configured lab, in parallel. A timeout abstains.

A hold goes to a human

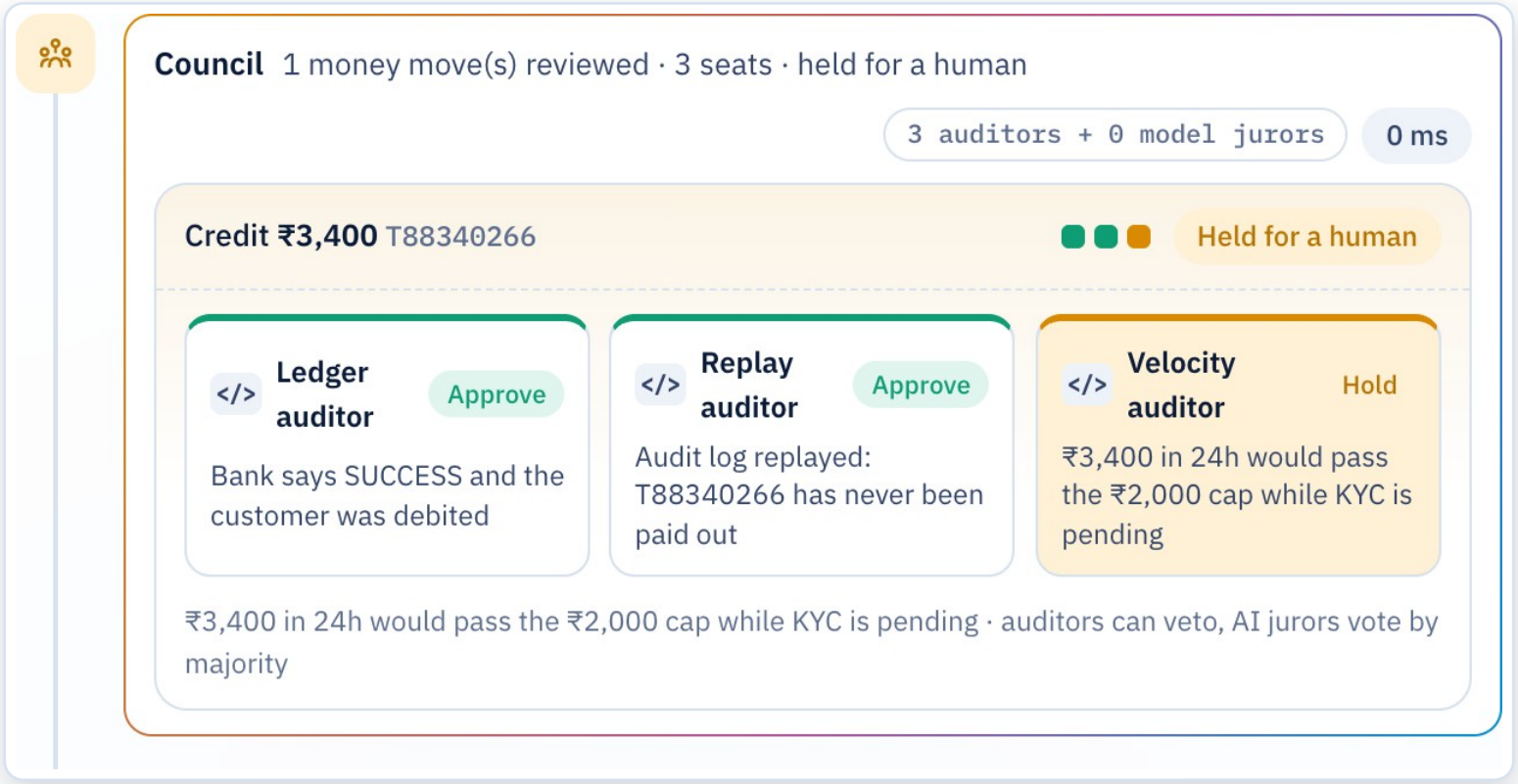
With every seat's reason. A human approval overrides.

Everything is logged

Each vote lands in the audit log.

Guardrails ask "is this allowed?" The council asks "is this safe right now?"

Demo 4: the council catches what the limit missed



1

₹3,400 is under the ₹5,000 limit

The guardrail allows it. A single-check system would pay out.

2

The velocity auditor objects

Gupta Medicals has KYC pending, so the autonomous cap is ₹2,000 a day.

3

Held, briefed, approved

The merchant gets a reply in Hindi; a specialist approves in one click.

Every major model, each on the job it’s best at

Agent	Job	Routing (first available wins)
Triage	Classify fast, pull out IDs	Groq Llama 3.1 8B → Gemini Flash → GPT-4.1 mini → Claude Haiku
Action	Plan and call tools in a loop	Claude Sonnet → GPT-4.1 → Gemini Pro → Groq Llama 70B
Reply	Warm Hindi / Hinglish / English	Gemini Flash → GPT-4.1 mini → Claude Haiku
Escalation	Two-sentence briefing	Claude Haiku → GPT-4.1 mini → Gemini Flash
Council	Independent votes on money	One juror per configured provider, all at once

6 providers: OpenAI, Anthropic, Gemini, Groq, OpenRouter, Ollama

2 adapters cover them all (OpenAI-compatible + Anthropic)

0 keys needed to demo: the rules engine is the safety net

Routing is configuration, so Paytm can swap in its own models without touching the agents.

Guardrails live in code, not in the prompt

Knows its limits

Below 60% confidence it escalates instead of guessing

Money limit

Credits and refunds above ₹5,000 need a human

Evidence first

Credit only after the bank confirms

Model Council

Auditors + cross-lab jurors on every money move

Stays in scope

Only the merchant it is serving

Grounded replies

Every ₹ and ID must match tool results

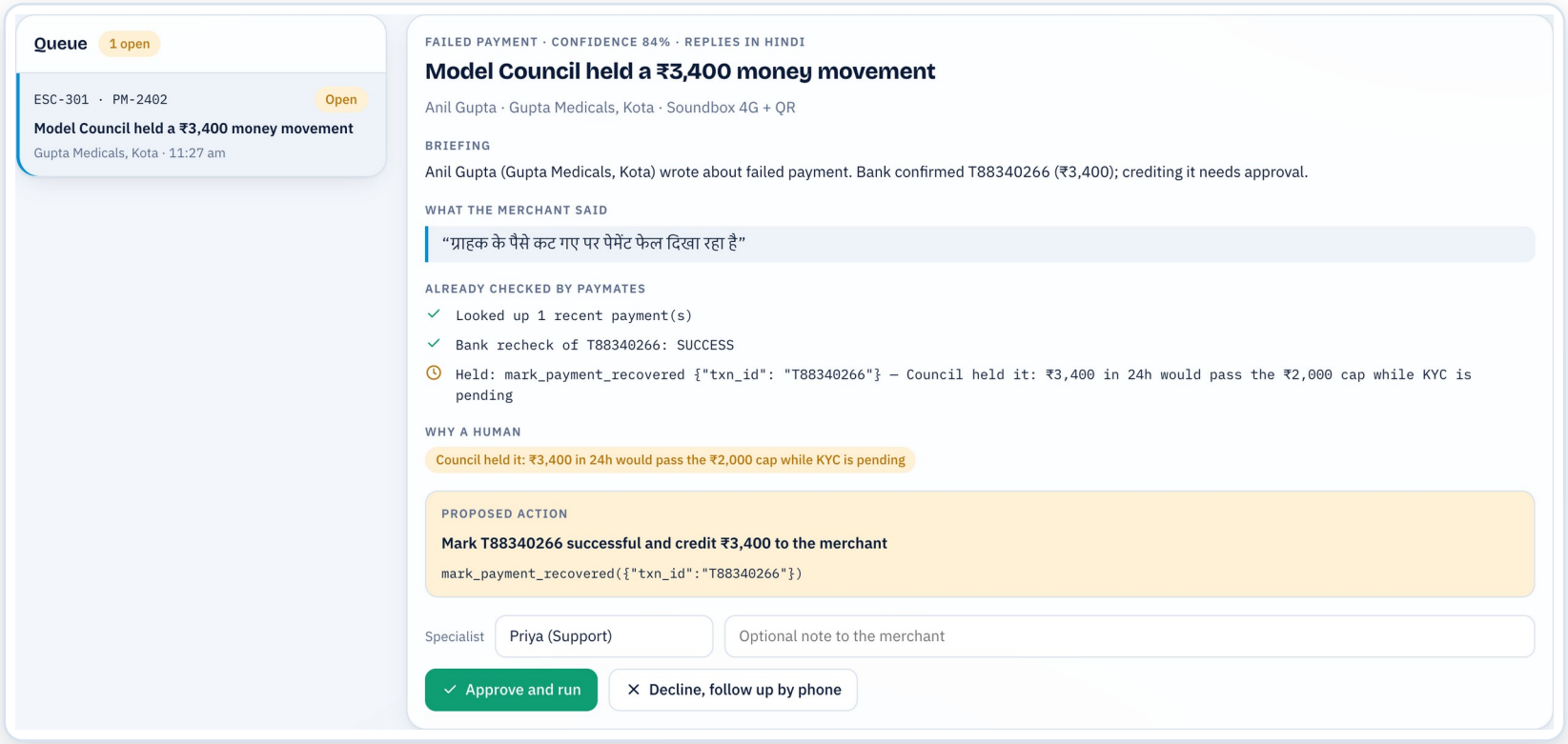
Private by default

Phones, UPI IDs, accounts masked before models

Full audit log

Who acted, and which rule allowed it

When a human is needed, they get a briefing, not a ticket



What the merchant said

In their own words and language

Already checked

Every tool call PayMates made, with results

Why a human is needed

The rule or council seat that held it

The exact action proposed

Approve and run executes it and messages the merchant

Measured on outcomes, not messages

Resolved without a human

82%

22 of 27 tickets

Resolved tickets ÷ all tickets

Median time to resolution

< 1 s

vs ~18 min by hand (assumed baseline)

Median of AI-resolved tickets, first message to closed, excluding on-screen pacing

Failed-payment value recovered

₹5,249

Bank-confirmed payments credited back to merchants

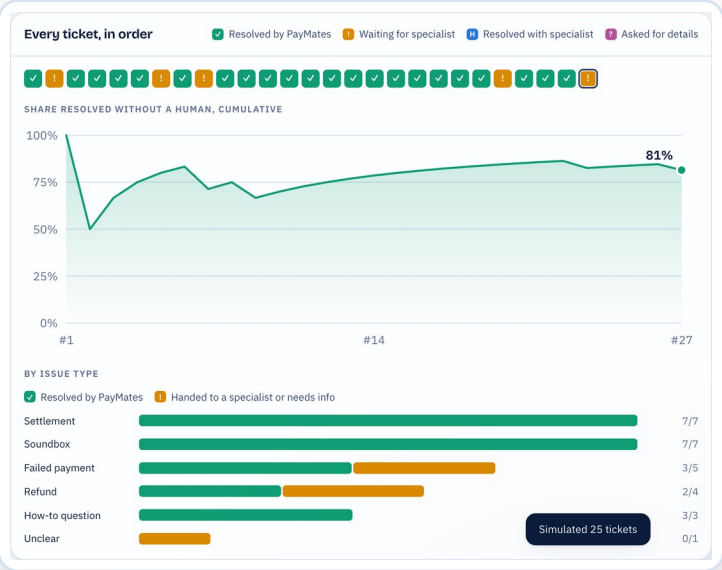
Sum of payments marked successful after a bank recheck

Specialist hours saved

6.6 h

5 waiting · 0 resolved with a specialist

AI-resolved tickets × 18 min average handling time



Simulated traffic

- 25 tickets, six merchants, three languages, through the same agents, council and guardrails
- All data is simulated; the method carries over to real tickets
- The 18-minute manual baseline is an assumption, and labelled as one

What we built it with

Agents & API

Python 3, FastAPI, async orchestrator, Server-Sent Events

Models

OpenAI, Claude, Gemini, Groq, OpenRouter, Ollama; native tool calling

Model Council

asyncio.gather jury, code auditors, abstain on timeout

RAG

TF-IDF vectors + cosine similarity; Hindi/Hinglish synonym expansion

Data

SQLite with plain SQL, ready to port to PostgreSQL

Web app

React 18, Vite, hand-built SVG charts, Web Speech API voice

Prototype

Same agent logic in JavaScript, runs fully in the browser

Quality

19 pytest end-to-end tests: demos, guardrails, council, fallbacks

From prototype to Paytm in three steps

01



Connect the tools

Tools already mirror merchant-support APIs: transactions, settlements, refunds, KYC, devices. Point them at Paytm's internal services.

02



Route Paytm's models

Model routing is configuration. Swap in in-house or preferred models per agent and per council seat.

03



Rerun the metrics on real tickets

Resolution rate, time to resolve, value recovered and hours saved, computed the same way.

THANK YOU

Support that feels like a teammate

Not a ticket number. Let’s get Raju home on time.

